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Project No.: 6

Project 6: Create EC2 Instances in Public and Private Subnets and Test Connectivity

Aim

To launch one Amazon EC2 instance in a Public Subnet and another Amazon EC2 instance in a Private Subnet within the same VPC, and verify network connectivity between them using their private IP addresses.

Prerequisites

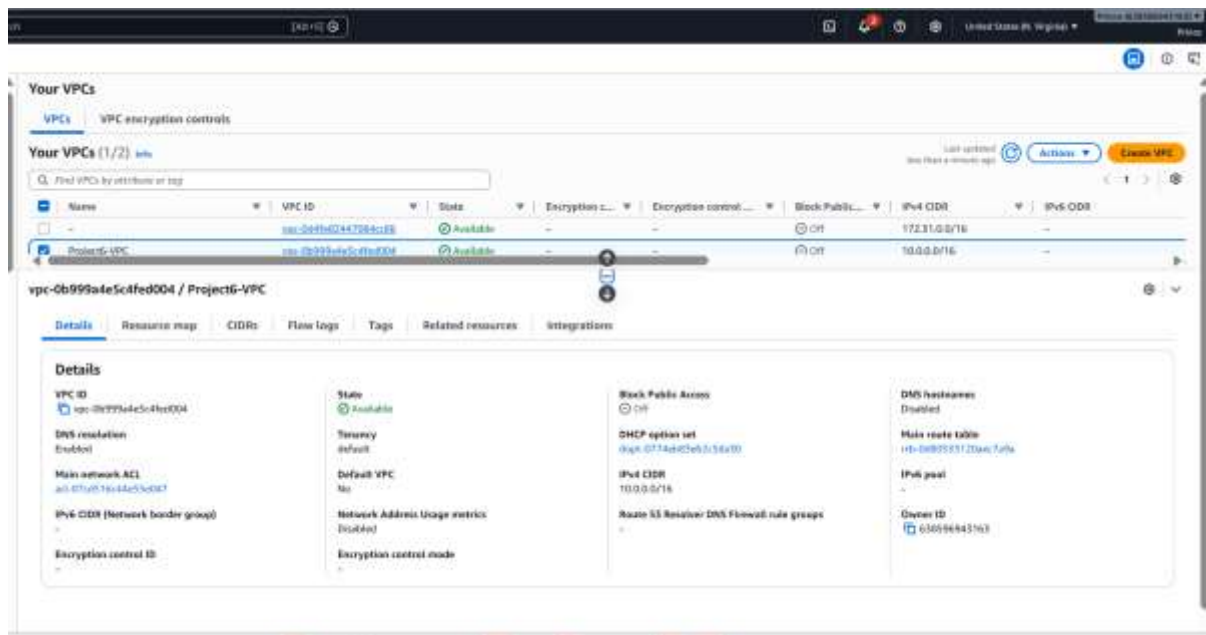
- AWS Account
 - IAM User with EC2 and VPC permissions or Root User
 - Custom VPC
 - Public Subnet and Private Subnet
 - Internet Gateway
 - Route Table for Public Subnet
 - Amazon Linux AMI
 - EC2 Key Pair
-

Procedure

Step 1: Create and Verify the VPC

1. Sign in to the AWS Management Console.
2. Navigate to **VPC**.
3. Create/select the custom VPC.
4. VPC CIDR block:
10.0.0.0/16

5. Verify that the VPC is available.



Step 2: Create Public and Private Subnets

Public Subnet

1. Navigate to **VPC → Subnets**.
2. Create a subnet inside the custom VPC.
3. Subnet Name: **Project6-Public-Subnet**
4. IPv4 CIDR:
10.0.1.0/24

Private Subnet

1. Create another subnet inside the same VPC.
2. Subnet Name: **Project6-Private-Subnet**
3. IPv4 CIDR:
10.0.2.0/24

The screenshot displays the AWS Management Console interface for the 'Subnets' section. The top navigation bar shows the user is logged in as 'arn:aws:iam::1:root' in the 'us-east-1' region. The 'Subnets' page shows a list of subnets, with 'subnet-00372ed0f48ff52' selected. The details for this subnet are shown below the list.

Name	Subnet ID	State	VPC	Block Public Access	IPv4 CIDR	IPv6 CIDR	IPv6
Project6-Public-Subnet	subnet-00372ed0f48ff52	Available	vpc-0b9934e5:us-east-1:1:33559043103:subnet/subnet-00372ed0f48ff52	Off	10.0.1.0/24	-	-
-	subnet-0d446d2447084c06	Available	vpc-0b9934e5:us-east-1:1:33559043103:subnet/subnet-0d446d2447084c06	Off	172.31.0.0/20	-	-
-	subnet-016773d061a5c52	Available	vpc-0b9934e5:us-east-1:1:33559043103:subnet/subnet-016773d061a5c52	Off	172.31.48.0/20	-	-
-	subnet-07a3c1a0a0c0d11	Available	vpc-0b9934e5:us-east-1:1:33559043103:subnet/subnet-07a3c1a0a0c0d11	Off	172.31.32.0/20	-	-

subnet-00372ed0f48ff52 / Project6-Public-Subnet

Details

- Subnet ID:** subnet-00372ed0f48ff52
- IPv4 CIDR:** 10.0.1.0/24
- Availability Zone:** us-east-1a
- Network ACL:** acl-07a3c1a0a0c0d11
- Auto-assign customer-owned IPv4 address:** No
- IPv6 CIDR reservations:** -
- Resource name DNS AAAA record:** Disabled

Subnet ARN: arn:aws:ec2:us-east-1:1:33559043103:subnet/subnet-00372ed0f48ff52

Available IPv4 addresses: 250

Network border group: bg-0000000000000000

Default subnet: No

Customer-owned IPv4 pool: -

IPv6-only: No

DNS64: Disabled

State: Available

IPv6 CIDR: -

VPC: vpc-0b9934e5:us-east-1:1:33559043103:Project6-VPC

Auto-assign public IPv4 address: No

Outpost ID: -

Hostname type: IP name

Owner: 638596943163

Block Public Access: Off

IPv6 CIDR association ID: -

Route table: rtr-0d446d2447084c06 / Project6-Public-RT

Auto-assign IPv6 address: No

IPv4 CIDR reservations: -

Resource name DNS A record: Disabled

The screenshot displays the AWS Management Console interface for the 'Subnets' section. The top navigation bar shows the user is logged in as 'arn:aws:iam::1:root' in the 'us-east-1' region. The 'Subnets' page shows a list of subnets, with 'subnet-03f34d5278ff1a382' selected. The details for this subnet are shown below the list.

Name	Subnet ID	State	VPC	Block Public Access	IPv4 CIDR	IPv6 CIDR	IPv6
Project6-Private-Subnet	subnet-03f34d5278ff1a382	Available	vpc-0b9934e5:us-east-1:1:33559043103:Project6-VPC	Off	10.0.2.0/24	-	-
-	subnet-0d446d2447084c06	Available	vpc-0b9934e5:us-east-1:1:33559043103:Project6-VPC	Off	172.31.48.0/20	-	-
-	subnet-07a3c1a0a0c0d11	Available	vpc-0b9934e5:us-east-1:1:33559043103:Project6-VPC	Off	172.31.32.0/20	-	-

subnet-03f34d5278ff1a382 / Project6-Private-Subnet

Details

- Subnet ID:** subnet-03f34d5278ff1a382
- IPv4 CIDR:** 10.0.2.0/24
- Availability Zone:** us-east-1a
- Network ACL:** acl-07a3c1a0a0c0d11
- Auto-assign customer-owned IPv4 address:** No
- IPv6 CIDR reservations:** -
- Resource name DNS AAAA record:** Disabled

Subnet ARN: arn:aws:ec2:us-east-1:1:33559043103:subnet/subnet-03f34d5278ff1a382

Available IPv4 addresses: 250

Network border group: bg-0000000000000000

Default subnet: No

Customer-owned IPv4 pool: -

IPv6-only: No

DNS64: Disabled

State: Available

IPv6 CIDR: -

VPC: vpc-0b9934e5:us-east-1:1:33559043103:Project6-VPC

Auto-assign public IPv4 address: No

Outpost ID: -

Hostname type: IP name

Owner: 638596943163

Block Public Access: Off

IPv6 CIDR association ID: -

Route table: rtr-0d446d2447084c06 / Project6-Private-RT

Auto-assign IPv6 address: No

IPv4 CIDR reservations: -

Resource name DNS A record: Disabled

Step 3: Configure Internet Gateway and Route Table

1. Navigate to **VPC → Internet Gateways**.
2. Create an Internet Gateway named **Project6-IGW**.

3. Attach the Internet Gateway to the custom VPC.

4. Create/select the Public Route Table.

5. Add the following route:

Destination: 0.0.0.0/0

Target: Internet Gateway

6. Associate the Public Subnet with the Public Route Table.

7. Associate the Private Subnet with a separate Private Route Table.

8. Do not add an Internet Gateway route to the Private Route Table.

Internet gateways (1/2)

Name	Internet gateway ID	State	VPC ID	Owner
Project5-IGW	igw-01508098d489b1634	Attached	vpc-0b07094c3e49b004 / Project5-VPC	638596943163
-	igw-Sec386884c5708af3	Attached	vpc-0b07094c3e49b004	638596943163

igw-01508098d489b1634 / Project5-IGW

Details

Internet gateway ID: igw-01508098d489b1634

State: Attached

VPC ID: vpc-0b07094c3e49b004 / Project5-VPC

Owner ID: 638596943163

Route tables (1/4)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
Project5-Public-RT	rtb-043773252abe831cb	subnet-003773ad094489f...	-	No	vpc-0b07094c3e49b004 / Proje...	638596943163
-	rtb-0b0f11371e45158a	-	-	No	vpc-0b07094c3e49b004 / Proje...	638596943163
-	rtb-0b0f0033120a57a8a	-	-	No	vpc-0b07094c3e49b004 / Proje...	638596943163
Project5-Private-RT	rtb-0b0f0033120a57a8a	subnet-003773ad094489f...	-	No	vpc-0b07094c3e49b004 / Proje...	638596943163

rtb-043773252abe831cb / Project5-Public-RT

Details

Route table ID: rtb-043773252abe831cb

VPC: vpc-0b07094c3e49b004 / Project5-VPC

Main: No

Owner ID: 638596943163

Explicit subnet associations: subnet-003773ad094489f... / Project5-Public-Subnet

Edge associations: -

Search

AMN

United States (N. Virginia)

Project: 00000000000000000000

rtb-045773232b6d011c6 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Project6-Public-Subnet	subnet-0073445278f1a382	10.0.1.0/24	-	rtb-045773232b6d011c6 / Project6-Pu...
Project6-Private-Subnet	subnet-03f44f5278f1a382	10.0.2.0/24	-	rtb-09865c0fc61e6e68c / Project6-Priv...

Selected subnets

subnet-0073445278f1a382 / Project6-Public-Subnet

Cancel Save associations

Search

AMN

United States (N. Virginia)

Project: 00000000000000000000

Route tables (1/4)

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet assoc...	Edge associations	Main	VPC	Owner ID
Project6-Public-RT	rtb-045773232b6d011c6	subnet-0073445278f1a382	-	No	vpc-08896a6316c1 / Project6-VPC	638590943163
-	rtb-045773232b6d011c6	-	-	Yes	vpc-08896a6316c1 / Project6-VPC	638590943163
-	rtb-09865c0fc61e6e68c	-	-	Yes	vpc-08896a6316c1 / Project6-VPC	638590943163
Project6-Private-RT	rtb-09865c0fc61e6e68c	subnet-03f44f5278f1a382	-	No	vpc-08896a6316c1 / Project6-VPC	638590943163

rtb-09865c0fc61e6e68c / Project6-Private-RT

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Details

Route table ID

rtb-09865c0fc61e6e68c

Main

No

Explicit subnet association

subnet-03f44f5278f1a382 / Project6-Private-Subnet

Edge associations

-

VPC

vpc-08896a6316c1 / Project6-VPC

Owner ID

638590943163

Search

AMN

United States (N. Virginia)

Project: 00000000000000000000

rtb-09865c0fc61e6e68c > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Project6-Public-Subnet	subnet-0073445278f1a382	10.0.1.0/24	-	rtb-045773232b6d011c6 / Project6-Pu...
Project6-Private-Subnet	subnet-03f44f5278f1a382	10.0.2.0/24	-	rtb-09865c0fc61e6e68c / Project6-Priv...

Selected subnets

subnet-03f44f5278f1a382 / Project6-Private-Subnet

Cancel Save associations

Step 4: Configure Security Groups

Public EC2 Security Group

Allow:

Type: SSH

Port: 22

Source: My IP

Private EC2 Security Group

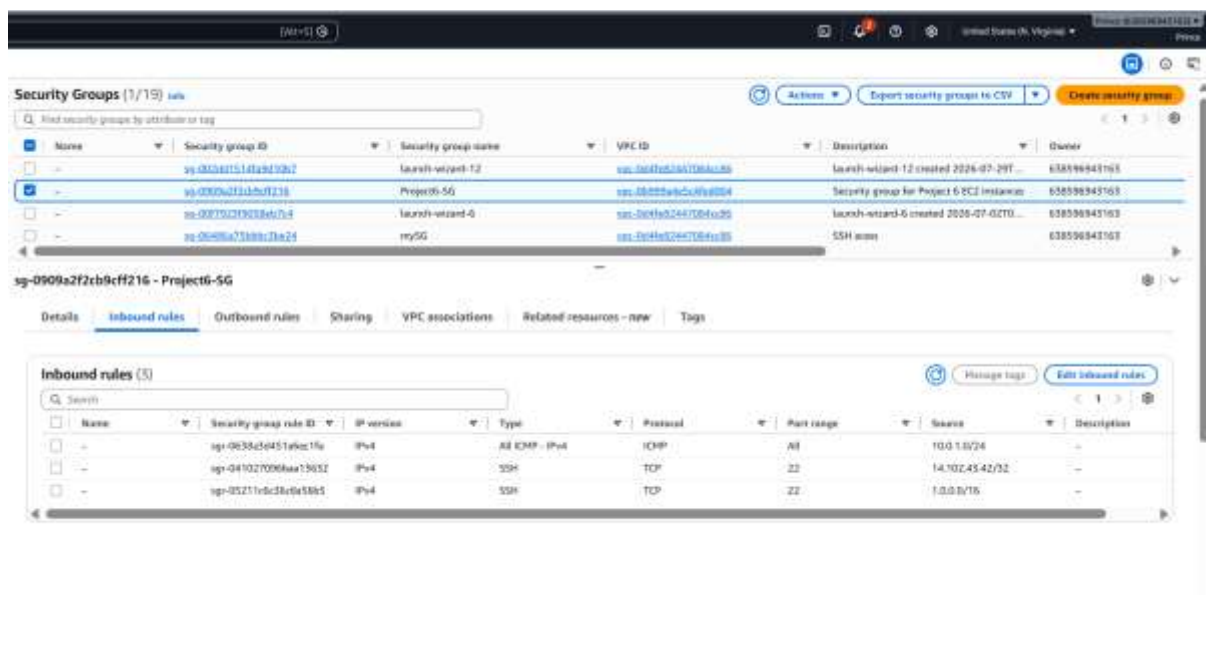
Allow:

Type: All ICMP - IPv4

Source: 10.0.1.0/24

This allows the Public EC2 in the 10.0.1.0/24 subnet to send ping requests to the Private EC2.

For SSH testing, SSH access can be restricted to the Public EC2/security group instead of opening SSH to the entire Internet.



Step 5: Launch an EC2 Instance in the Public Subnet

1. Navigate to **EC2 → Instances**.
2. Click **Launch Instance**.
3. Enter the following details:

Instance Name:

Project6-Public-EC2

AMI:

Amazon Linux

Instance Type:

Select the available suitable instance type.

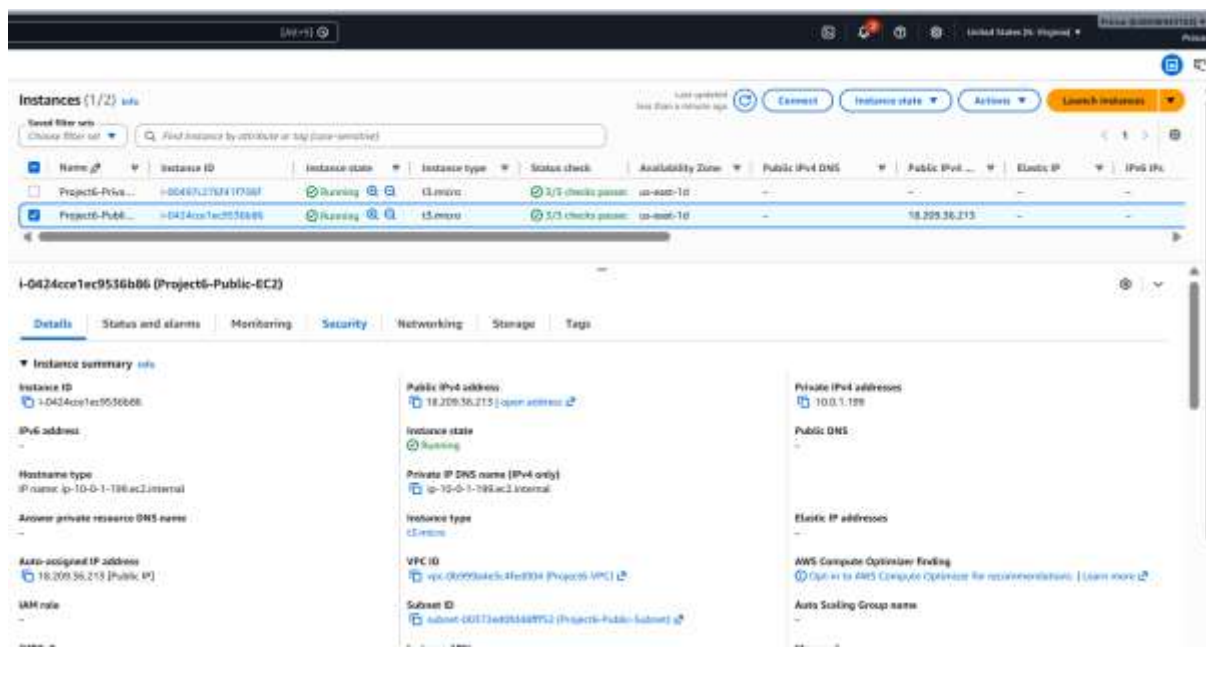
Key Pair:

Select the existing EC2 key pair.

4. Under **Network Settings:**

- VPC: **Project6-VPC**
- Subnet: **Project6-Public-Subnet**
- Auto-assign Public IP: **Enable**
- Security Group: **Project6-Public-SG**

5. Launch the instance.

**Step 6: Launch an EC2 Instance in the Private Subnet**

1. Click **Launch Instance** again.
2. Enter:

Instance Name:

Project6-Private-EC2

AMI:

Amazon Linux

Instance Type:

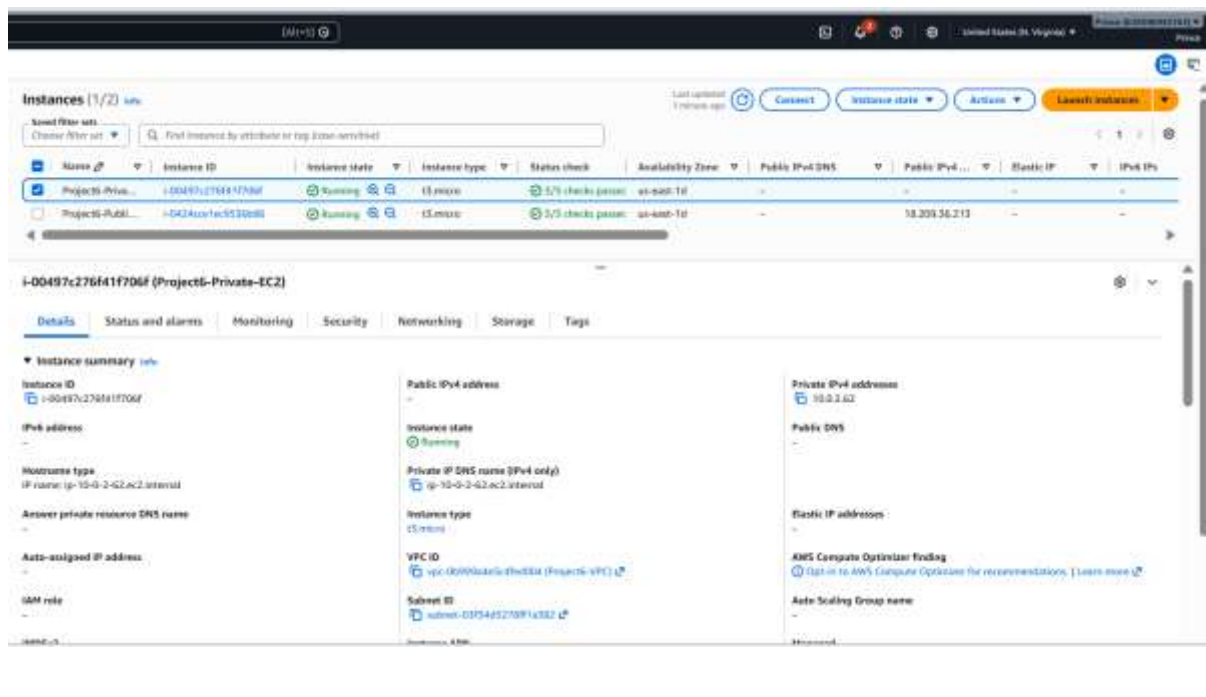
Select the suitable available instance type.

Key Pair:

Select the same key pair.

3. Under Network Settings:

- VPC: **Project6-VPC**
- Subnet: **Project6-Private-Subnet**
- Auto-assign Public IP: **Disable**
- Security Group: **Project6-Private-SG**

4. Launch instance**Step 7: Verify Both EC2 Instances**

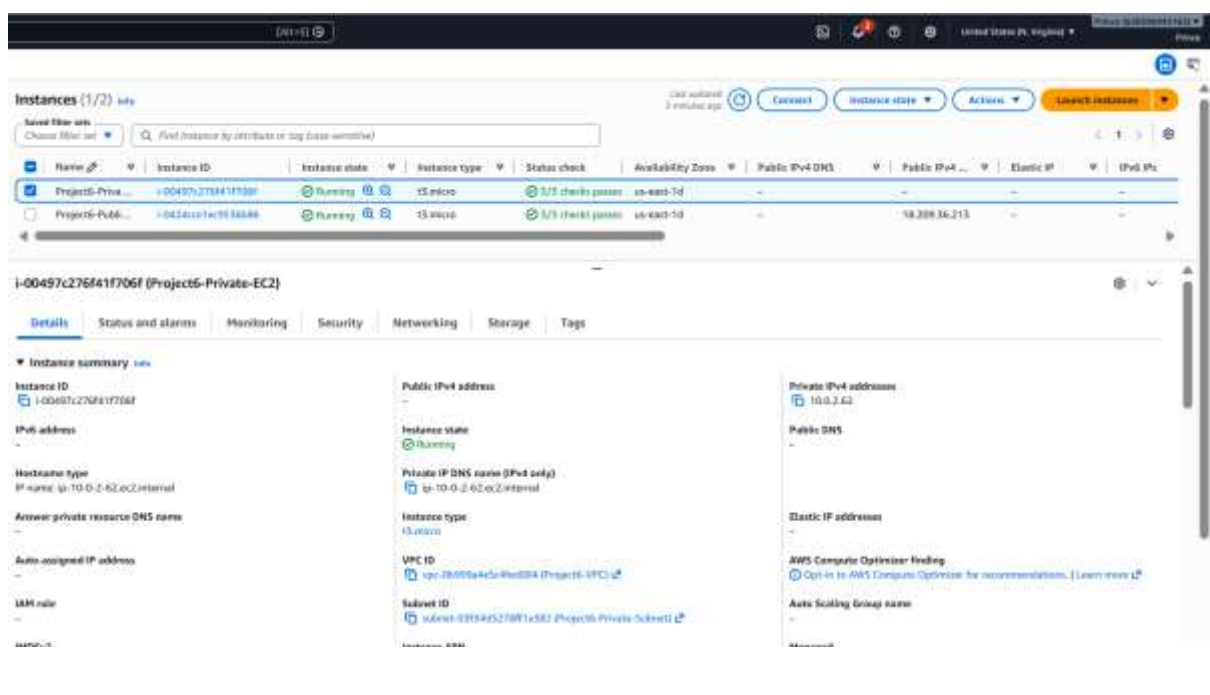
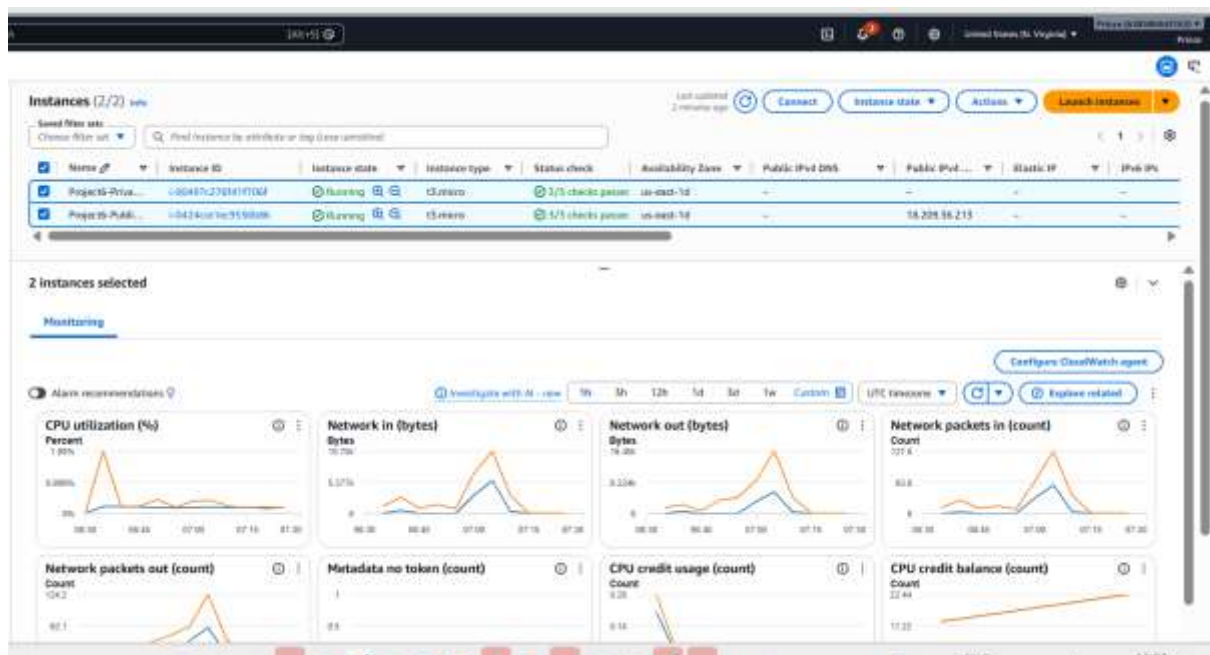
1. Open **EC2** → **Instances**.
2. Verify that both instances are in the **Running** state.
3. Note their Private IPv4 addresses.

4. Verify that the Public EC2 has a Public IPv4 address.
5. Verify that the Private EC2 does not have a Public IPv4 address.

For this project, our instances were:

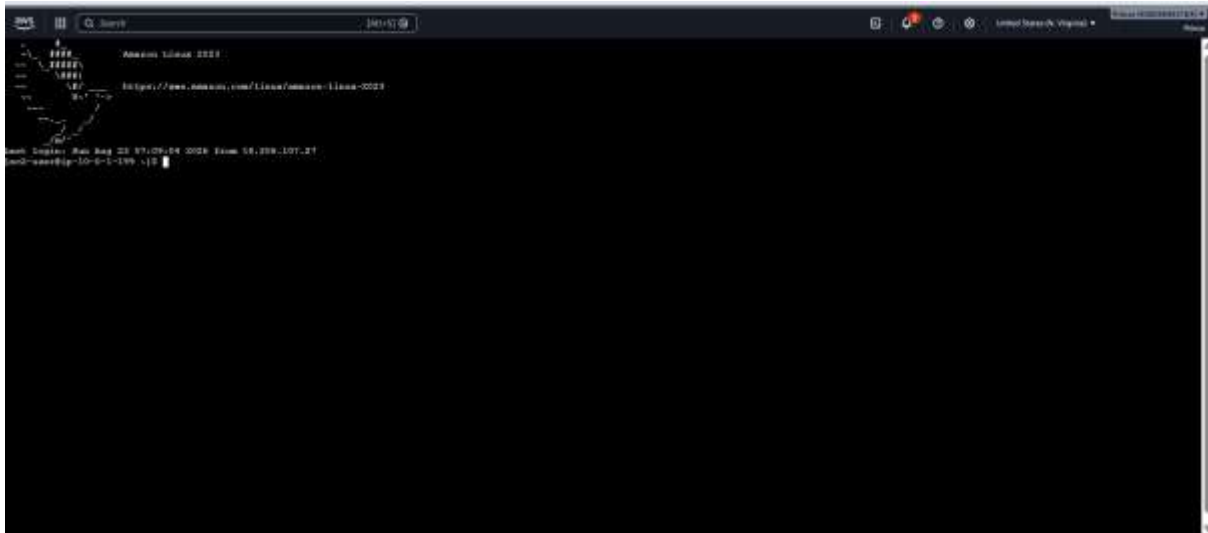
Public EC2 Private IP: 10.0.1.83

Private EC2 Private IP: 10.0.2.62



Step 8: Connect to the Public EC2 Instance

- Successful login confirms that the Public EC2 is reachable.



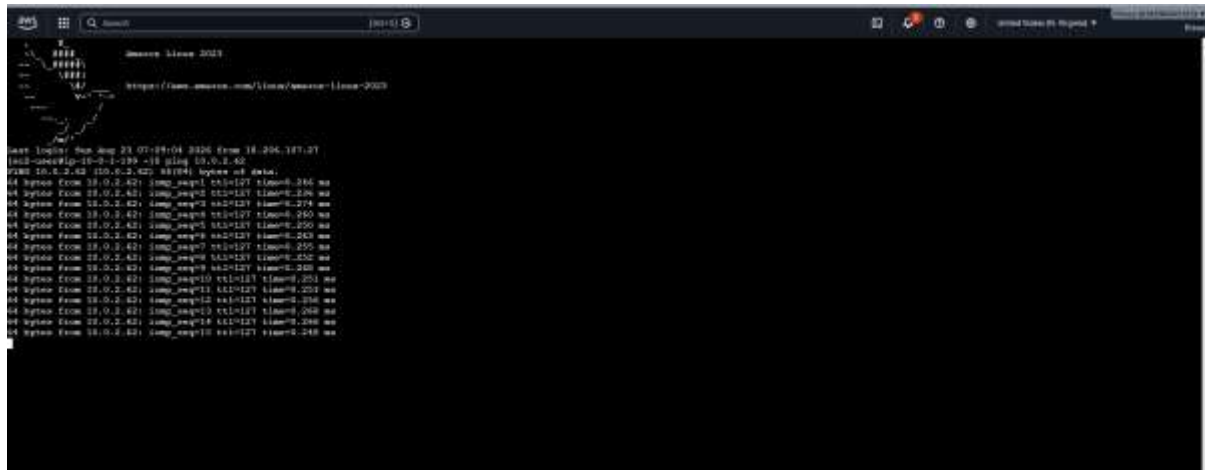
From the Public EC2 terminal, run:

The Private EC2 should respond with messages similar to:

64 bytes from 10.0.2.62

64 bytes from 10.0.2.62

This confirms that the Public EC2 can communicate with the Private EC2 using its **Private IP address**.



```
root@ip-10-0-2-42:~# ping -c 1 10.0.2.42
PING 10.0.2.42 (10.0.2.42) 64(84) bytes of data:
64 bytes from 10.0.2.42: icmp_seq=1 ttl=64 time=0.260 ms
64 bytes from 10.0.2.42: icmp_seq=2 ttl=64 time=0.256 ms
64 bytes from 10.0.2.42: icmp_seq=3 ttl=64 time=0.274 ms
64 bytes from 10.0.2.42: icmp_seq=4 ttl=64 time=0.260 ms
64 bytes from 10.0.2.42: icmp_seq=5 ttl=64 time=0.250 ms
64 bytes from 10.0.2.42: icmp_seq=6 ttl=64 time=0.263 ms
64 bytes from 10.0.2.42: icmp_seq=7 ttl=64 time=0.257 ms
64 bytes from 10.0.2.42: icmp_seq=8 ttl=64 time=0.252 ms
64 bytes from 10.0.2.42: icmp_seq=9 ttl=64 time=0.240 ms
64 bytes from 10.0.2.42: icmp_seq=10 ttl=64 time=0.263 ms
64 bytes from 10.0.2.42: icmp_seq=11 ttl=64 time=0.253 ms
64 bytes from 10.0.2.42: icmp_seq=12 ttl=64 time=0.256 ms
64 bytes from 10.0.2.42: icmp_seq=13 ttl=64 time=0.260 ms
64 bytes from 10.0.2.42: icmp_seq=14 ttl=64 time=0.266 ms
64 bytes from 10.0.2.42: icmp_seq=15 ttl=64 time=0.248 ms
```

Step 10: Verify Network Configuration

Verify the following:

- Both EC2 instances are in the **same VPC**.
- Public EC2 is in the **Public Subnet**.
- Private EC2 is in the **Private Subnet**.
- Public EC2 has a **Public IPv4 address**.
- Private EC2 has **no Public IPv4 address**.
- Public Route Table has:
0.0.0.0/0 → Internet Gateway
- Public EC2 can communicate with Private EC2 using its Private IP.
- Security Group allows ICMP between the required subnet/instances.

Instances (1/2)

Save filter sets

Find instances by attribute or tag (case-sensitive)

Choose filter set

Find instances by attribute or tag (case-sensitive)

Connect

Instance state

Actions

Search instances

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IPs
Project6-Priv...	i-00497c276f41f706f	Running	t3.micro	3/3 checks passed	us-east-1d	-	-	-	-
Project6-Publ...	i-0424cce1ec9556b86	Running	t3.micro	3/3 checks passed	us-east-1d	-	18.209.36.213	-	-

I-00497c276f41f706f (Project6-Private-EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID

apc-08889a6c3af8a504 (Project6-VPC)

Subnet ID

subnet-03f3ad278f1a102 (Project6-Private-Subnet)

Availability zone

us-east-1d

Availability zone ID

us-east-1a

Outpost ID

-

IP addresses

Public IPv4 address

Private IPv4 address

IPv6 address

Secondary private IPv4 addresses

Carrier IP address (ephemeral)

Hostnames and DNS

Public DNS

Private IP DNS name (IPv4 only)

IPv4-only IP based name: A record only

Dualstack - IP based name: A and AAAA record

IPv6-only - IP based name: AAAA record only

Public instance type

Instances (1/2)

Save filter sets

Find instances by attribute or tag (case-sensitive)

Choose filter set

Find instances by attribute or tag (case-sensitive)

Connect

Instance state

Actions

Search instances

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IPs
Project6-Priv...	i-00497c276f41f706f	Running	t3.micro	3/3 checks passed	us-east-1d	-	-	-	-
Project6-Publ...	i-0424cce1ec9556b86	Running	t3.micro	3/3 checks passed	us-east-1d	-	18.209.36.213	-	-

i-0424cce1ec9556b86 (Project6-Public-EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

Public IPv4 address

Private IPv4 address

IPv6 address

Instance state

Public DNS

Instance type

Private IP DNS name (IPv4 only)

Elastic IP address

Instance private resource DNS name

Instance type

Amazon Compute Optimizer finding

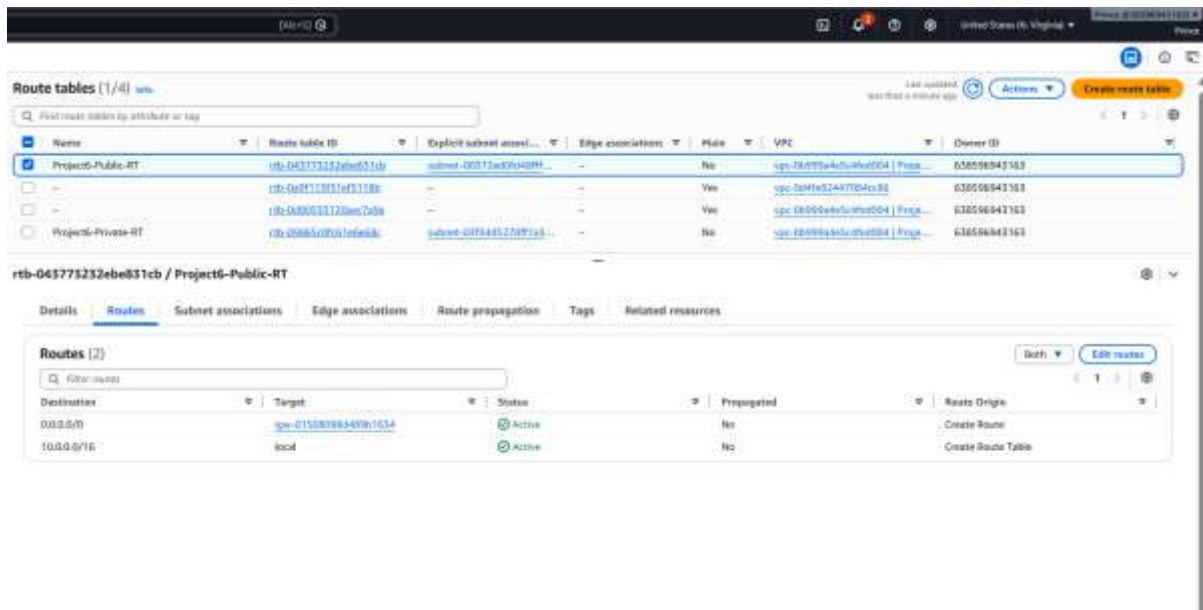
Auto-assigned IP address

VPC ID

Auto Scaling Group name

AMI role

Subnet ID



Result

The Public EC2 and Private EC2 instances were successfully launched in their respective subnets within the same VPC. The Public EC2 instance was successfully connected using SSH, and connectivity between the Public and Private EC2 instances was successfully verified using the Private IP address.

Conclusion

This project demonstrated how to deploy EC2 instances in Public and Private Subnets within a custom VPC. The Public EC2 instance was configured with Internet connectivity through an Internet Gateway, while the Private EC2 instance did not have a Public IP address. Connectivity between the two instances was successfully verified using their Private IP addresses.